

TYPE I BLOW-UP LIMITS FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS AND THE CENTERED ANCIENT EQUATION

ABSTRACT. We give a detailed reduction from a Type I singular point of a suitable Leray–Hopf solution of the three-dimensional incompressible Navier–Stokes equations to a bounded ancient suitable weak solution in centered self-similar variables. The construction records the precise scaling, pressure gauges, local compactness, local energy stability, and the pressure-normalized Caffarelli–Kohn–Nirenberg concentration which prevents the limit from vanishing. The compactness and nonvanishing steps use the Caffarelli–Kohn–Nirenberg–Seregin blow-up compactness mechanism in the precise form recorded below. The resulting renormalized ancient field solves

$$\partial_\tau V + (V \cdot \nabla)V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0,$$

on $\mathbb{R}^3 \times \mathbb{R}$, and is uniformly bounded in $L_{y,\tau}^\infty$. We also observe that this centered equation has no non-trivial positive scaling symmetry and is not invariant under constant spatial translations. Parabolic rescalings of the unrenormalized ancient solution appear only as translations in the logarithmic time variable. Thus centered Type I ancient limits are not organized by an additional dilation orbit in the centered variables.

1. INTRODUCTION

Let $u : \mathbb{R}^3 \times [0, T) \rightarrow \mathbb{R}^3$ solve the incompressible Navier–Stokes equations

$$(1.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \quad \operatorname{div} u = 0.$$

The local regularity theory of Caffarelli–Kohn–Nirenberg [3] shows that singular points are detected by scale-invariant concentration of velocity and pressure. A finite-time singular point $z_* = (x_*, T)$ is said to be of Type I if the scale-invariant L^∞ quantity remains bounded,

$$(1.2) \quad \limsup_{t \uparrow T} \sqrt{T-t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

For the endpoint compactness argument one also needs the scale-invariant energy and pressure quantities at cylinders ending at z_* to remain finite. We record this as the hypothesis $\mathcal{I}_*(u, p; z_*) < \infty$. This is the local compactness formulation of Type I used in the partial regularity literature; it follows from the standard critical Morrey or weak Serrin Type I hypotheses, and it is the condition used below to pass to a suitable limit up to the terminal slice. For the relation between local Type I hypotheses, scale-invariant energy control, and bounded ancient limits, see Albritton–Barker [1, Theorem 1.1]. The purpose of this paper is to isolate the canonical ancient object associated with such a singular point.

The pressure-normalized CKN quantities, their parabolic scaling, the epsilon-regularity/no-concentration implication, and the local Type I/Type II rate dichotomy were established in the

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first-node paper [8, Sections 2–5]. The specific internal labels used below are

$$\begin{aligned} &\text{prop:critical-scaling,} \quad \text{thm:ckn,} \\ &\text{thm:singular-positive,} \quad \text{thm:blowup-rate-dichotomy.} \end{aligned}$$

We recall the notation needed below, but we do not reprove those foundational facts. The additional work in the present paper is the Type I compactness up to ancient time, the pressure gauges needed for the whole-space limit, the nonvanishing normalization of the selected limit, and the centered self-similar equation.

The parabolic scaling around z_* is

$$(1.3) \quad u^{(\lambda)}(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^{(\lambda)}(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

As $\lambda \downarrow 0$, the time intervals $(-T/\lambda^2, 0)$ exhaust $(-\infty, 0)$. The Type I bound becomes

$$\|u^{(\lambda)}(t)\|_{L^\infty(\mathbb{R}^3)} \lesssim (-t)^{-1/2}, \quad t < 0,$$

uniformly on compact subintervals of $(-\infty, 0)$. This gives local compactness away from $t = 0$. After selecting scales carefully, one obtains a non-zero ancient suitable weak limit U .

The centered self-similar variables associated with the terminal time are

$$(1.4) \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t).$$

Then V solves the autonomous equation

$$(1.5) \quad \partial_\tau V + (V \cdot \nabla) V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0.$$

The drift term in (1.5) is the signature of the Type I scaling. It fixes a scale and center. Consequently, unlike the original Navier–Stokes equations, (1.5) has no non-trivial amplitude-space scaling symmetry and no constant translation symmetry in the y -variable. The remaining symmetry is translation in τ , which is the image of parabolic rescaling before the centered change of variables. Consequently, after the centered change of variables has been made, Type I ancient dynamics are not generated by further same-center parabolic dilations.

1.1. Main result. The following theorem is the main result of the paper. The non-triviality assertion is stated in the pressure-normalized CKN form supplied by the local concentration theorem at singular points.

Theorem 1.1 (Type I blow-up limit and centered ancient equation). *Let (u, p) be a suitable Leray–Hopf solution of (1.1) on $\mathbb{R}^3 \times (0, T)$. Assume $z_* = (x_*, T)$ is a singular point and that (1.2) holds. Assume in addition that the centered endpoint Type I compactness quantity is finite:*

$$(1.6) \quad \mathcal{I}_*(u, p; z_*) < \infty.$$

Here $\mathcal{I}_$ is defined in (2.6). Then there are scales $\lambda_k \downarrow 0$, functions of time $a_k(t)$, and an ancient suitable weak solution (U, P) on $\mathbb{R}^3 \times (-\infty, 0)$ such that, with*

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \quad \tilde{p}_k(x, t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t), \quad \pi_k = \tilde{p}_k - a_k(t),$$

one has, after passing to a subsequence,

$$u_k \rightarrow U \quad \text{strongly in } L_{\text{loc}}^q(\mathbb{R}^3 \times (-\infty, 0)) \quad \text{for every } 1 \leq q < \infty,$$

and

$$\pi_k \rightharpoonup P \quad \text{weakly in } L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (-\infty, 0)).$$

The limit satisfies

$$(1.7) \quad \|U(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}, \quad t < 0,$$

for a finite constant M depending only on the Type I bound. Moreover the scales may be chosen so that $U \not\equiv 0$. The renormalized field V defined by (1.4) satisfies

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M$$

and is a smooth solution of (1.5) on $\mathbb{R}^3 \times \mathbb{R}$. The non-triviality is quantified as follows: there are a compact cylinder

$$K \Subset \mathbb{R}^3 \times \mathbb{R},$$

a function $b(\tau)$, and $\eta_0 > 0$ such that

$$(1.8) \quad \iint_K \left(|V|^3 + |\Pi - b(\tau)|^{3/2} \right) dy d\tau \geq \eta_0.$$

Proposition 1.2 (No additional dilation or translation symmetry). *The centered equation (1.5) is autonomous in τ , but it is not invariant under the positive amplitude-space scaling*

$$V(y, \tau) \mapsto \alpha V(\alpha y, \alpha^2 \tau)$$

unless $\alpha = 1$. It is also not invariant under translations

$$V(y, \tau) \mapsto V(y - a, \tau)$$

unless $a = 0$. Hence the centered Type I equation has no intrinsic same-center parabolic dilation symmetry beyond translation of the logarithmic time variable.

1.2. Organization. Section 2 fixes notation and recalls suitable weak solutions, scale-invariant quantities, and epsilon regularity. Section 3 proves uniform bounds for the rescaled sequence. Section 4 passes to an ancient suitable weak limit. Section 5 proves the non-triviality normalization. Section 6 derives the centered equation. Section 7 proves Theorem 1.2.

1.3. Proof bookkeeping. The proof has three logically separate layers. First, the Type I bound gives local L^∞ compactness on every cylinder separated from the terminal time. Second, the finite endpoint compactness quantity $\mathcal{I}_*(u, p; z_*)$ gives compactness up to rescaled cylinders ending at $(0, 0)$, after subtracting time-dependent pressure gauges. Third, the positive pressure-normalized CKN concentration at the singular point chooses the scales so that the limiting ancient solution is not zero. None of these three layers implies the others. In particular, the Type I bound alone gives boundedness of possible limits but not non-triviality, while local concentration alone gives non-triviality but not the global-in-time L^∞ ancient bound.

2. PRELIMINARIES

We write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0), \quad z_0 = (x_0, t_0).$$

When $z_0 = (0, 0)$ we write $Q_r = Q_r(0, 0)$. The spatial average of a function f over $B_r(x_0)$ is denoted by

$$(f)_{B_r(x_0)}(t) = \frac{1}{|B_r|} \int_{B_r(x_0)} f(x, t) dx.$$

Definition 2.1 (Terminal singular point). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. A terminal point $z_* = (x_*, T)$ is called regular if there are $r > 0$ and $K < \infty$ such that

$$\operatorname{ess\,sup}_{Q_r(z_*)} |u| \leq K.$$

It is called singular otherwise.

Definition 2.2 (Suitable Leray–Hopf solution). Let $\Omega \subset \mathbb{R}^3 \times \mathbb{R}$ be open. A pair (u, p) is a suitable weak solution of (1.1) in Ω if

$$u \in L_t^\infty L_x^2(\Omega') \cap L_t^2 \dot{H}_x^1(\Omega'), \quad p \in L^{3/2}(\Omega')$$

for every compact cylinder $\Omega' \Subset \Omega$, if (u, p) solves (1.1) distributionally, $\operatorname{div} u = 0$, and if the local energy inequality

$$\begin{aligned} & \int_{\mathbb{R}^3} |u(x, t_2)|^2 \phi(x, t_2) dx + 2 \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi dx dt \\ & \leq \int_{\mathbb{R}^3} |u(x, t_1)|^2 \phi(x, t_1) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |u|^2 (\partial_t \phi + \Delta \phi) dx dt \\ & + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} (|u|^2 + 2p) u \cdot \nabla \phi dx dt \end{aligned} \quad (2.1)$$

holds for every non-negative $\phi \in C_c^\infty(\Omega)$ and almost every $t_1 < t_2$ for which the displayed terms are defined.

This is the Caffarelli–Kohn–Nirenberg notion of suitability [3]; the global finite-energy background is Leray’s Leray–Hopf class [6].

We use the standard scale-invariant quantities

$$(2.2) \quad A(u; z_0, r) = r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx,$$

$$(2.3) \quad C(u; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |u|^3 dx dt, \quad D(p; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

$$(2.4) \quad E(u; z_0, r) = r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 dx dt,$$

$$(2.5) \quad \mathcal{I}(u, p; z_0, r) = A(u; z_0, r) + C(u; z_0, r) + D(p; z_0, r) + E(u; z_0, r).$$

For a terminal point $z_* = (x_*, T)$, we write

$$(2.6) \quad \mathcal{I}_*(u, p; z_*) = \sup_{0 < r < r_*} \mathcal{I}(u, p; z_*, r),$$

where $r_* > 0$ is fixed sufficiently small that the corresponding cylinders lie in the spacetime domain under consideration. Finiteness of $\mathcal{I}_*(u, p; z_*)$ is independent of decreasing r_* .

Theorem 2.3 (Epsilon regularity). *There exists $\varepsilon_* > 0$ with the following property. If (u, p) is a suitable weak solution in $Q_r(z_0)$ and*

$$C(u; z_0, r) + D(p; z_0, r) \leq \varepsilon_*,$$

then u is bounded, and hence smooth, in $Q_{r/2}(z_0)$.

This is the Caffarelli–Kohn–Nirenberg regularity criterion [3]; see also Lin’s direct proof [5]. We shall use only its contrapositive.

Lemma 2.4 (Scaling). *Let (u, p) be a suitable weak solution in a neighborhood of (x_*, T) . For $\lambda > 0$ set*

$$u^\lambda(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^\lambda(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

Then (u^λ, p^λ) is a suitable weak solution on the rescaled domain. Moreover A, C, D, E in (2.2)–(2.4) are invariant under this scaling.

Proof. Suitability of the rescaled pair, invariance of C and D , and preservation of pressure gauges are the parabolic rescaling results proved in the first-node paper [8, Section 2 and Proposition `prop:critical-scaling`]. The remaining two quantities used here have the same critical dimensional balance. With $X = x_* + \lambda x$ and $S = T + \lambda^2 t$,

$$\begin{aligned} \int_{B_r} |u^\lambda(x, t)|^2 dx &= \lambda^{-1} \int_{B_{\lambda r}(x_*)} |u(X, S)|^2 dX, \\ \iint_{Q_r} |\nabla u^\lambda|^2 dx dt &= \lambda^{-1} \iint_{Q_{\lambda r}(z_*)} |\nabla u|^2 dX dS. \end{aligned}$$

Multiplication by r^{-1} on the rescaled cylinder is therefore the same as multiplication by $(\lambda r)^{-1}$ on the original cylinder. Hence A and E are invariant as well. $\square$

Lemma 2.5 (Pressure gauges do not change the equations). *Let (v, q) be a suitable weak solution in a cylinder Q , and let $\alpha(t) \in L_{\text{loc}}^{3/2}$ depend only on time. Then $(v, q - \alpha(t))$ satisfies the same distributional Navier–Stokes equation, the same divergence condition, and the same local energy inequality.*

Proof. In the distributional equation, replacing q by $q - \alpha(t)$ adds $-\nabla \alpha(t)$, which is zero because α has no spatial dependence. The divergence equation is independent of the pressure. In the local energy inequality the only new term is

$$-2 \int \alpha(t) v(x, t) \cdot \nabla \phi(x, t) dx dt.$$

For almost every t , since $\text{div } v(\cdot, t) = 0$ in distributions and $\phi(\cdot, t)$ is compactly supported,

$$\int v(x, t) \cdot \nabla \phi(x, t) dx = 0.$$

Thus the additional term vanishes. Therefore all compactness arguments may choose pressure representatives modulo functions of time without altering suitability. $\square$

3. UNIFORM ESTIMATES FOR THE RESCALED SEQUENCE

Let $M < \infty$ be chosen so that, after decreasing $T_0 < T$ if necessary,

$$(3.1) \quad \|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T-t}}, \quad T_0 < t < T.$$

For a sequence $\lambda_k \downarrow 0$, define

$$(3.2) \quad u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \quad p_k(x, t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then u_k is defined on

$$\mathbb{R}^3 \times (-T/\lambda_k^2, 0),$$

and these domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$.

Lemma 3.1 (Inherited Type I bound). *For every compact interval $I \Subset (-\infty, 0)$,*

$$\sup_k \|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} < \infty.$$

More precisely, for every fixed $t < 0$ and all sufficiently large k ,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}.$$

Proof. For $t < 0$,

$$T - (T + \lambda_k^2 t) = \lambda_k^2(-t).$$

Thus (3.1) gives, for all large k ,

$$\|u_k(t)\|_\infty = \lambda_k \|u(T + \lambda_k^2 t)\|_\infty \leq \lambda_k \frac{M}{\sqrt{\lambda_k^2(-t)}} = \frac{M}{\sqrt{-t}}.$$

The interval statement follows by taking the supremum over $t \in I$. $\square$

Lemma 3.2 (Local energy and dissipation). *Let*

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \quad 0 < \sigma < S < \infty.$$

Then

$$\sup_k \|u_k\|_{L_t^\infty L_x^2(Q)} + \sup_k \|\nabla u_k\|_{L^2(Q)} < \infty.$$

Proof. The $L_t^\infty L_x^2$ estimate is a direct consequence of Theorem 3.1:

$$\|u_k(t)\|_{L^2(B_R)} \leq |B_R|^{1/2} \frac{M}{\sqrt{\sigma}}, \quad t \in [-S, -\sigma].$$

To estimate ∇u_k , choose $\phi \in C_c^\infty(B_{2R} \times [-2S, -\sigma/2])$, $0 \leq \phi \leq 1$, with $\phi \equiv 1$ on Q . Apply the local energy inequality to $(u_k, p_k - a_k(t))$, where $a_k(t) = (p_k)_{B_1}(t)$ is the fixed reference gauge from Theorem 3.3. The estimate in Theorem 3.3 is used on the larger cylinder $B_{2R} \times [-2S, -\sigma/2]$. Subtracting this function of time from the pressure does not change the local energy inequality by Theorem 2.5. For a.e. admissible pair of times $-2S < t_1 < t_2 < -\sigma/2$, and then by monotone approximation of the temporal cutoff to the endpoints of the support of ϕ , the inequality gives

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 |\partial_t \phi + \Delta \phi| + \iint (|u_k|^2 + 2|p_k - a_k(t)|) |u_k| |\nabla \phi|.$$

The first term is bounded by the inherited L^∞ bound and the finite measure of $B_{2R} \times [-2S, -\sigma/2]$. For the cubic velocity term,

$$\iint |u_k|^3 |\nabla \phi| \leq \|\nabla \phi\|_\infty |B_{2R} \times [-2S, -\sigma/2]| \|u_k\|_{L^\infty(B_{2R} \times [-2S, -\sigma/2])}^3.$$

For the pressure term, Hölder's inequality with exponents $3/2$ and 3 gives

$$\iint |p_k - a_k(t)| |u_k| |\nabla \phi| \leq \|p_k - a_k(t)\|_{L^{3/2}} \|u_k \nabla \phi\|_{L^3},$$

and both factors are uniformly bounded by Theorem 3.3 and Theorem 3.1. Absorbing no term is necessary, since the dissipation appears with a favorable sign. Because $\phi \equiv 1$ on Q , this yields

$$\iint_Q |\nabla u_k|^2 dx dt \leq C(R, S, \sigma, M).$$

$\square$

Lemma 3.3 (Pressure gauges and local pressure bounds). *After fixing the whole-space Leray pressure representative below, set*

$$a_k(t) = (p_k)_{B_1}(t)$$

whenever the rescaled solution is defined. Then for every $Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$,

$$\sup_k \|p_k - a_k(t)\|_{L^{3/2}(Q)} < \infty.$$

Proof. For all sufficiently large k , the original times $T + \lambda_k^2 t$, $t \in [-S, -\sigma]$, lie in the interval (T_0, T) , where the Type I bound (3.1) holds. On those time slices $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence

$$u_i(t)u_j(t) \in L^1(\mathbb{R}^3) \cap L^{3/2}(\mathbb{R}^3)$$

by interpolation. We use the whole-space Leray pressure representative

$$p^L = \mathcal{R}_i \mathcal{R}_j (u_i u_j),$$

modulo functions of time; this is the pressure reconstruction from $-\Delta p = \partial_i \partial_j (u_i u_j)$ and the Calderon–Zygmund theory of Riesz transforms [13, 12]. For whole-space Leray–Hopf solutions the given pressure and p^L differ only by a function of time. Indeed, both pressures give the same gradient part in the distributional momentum equation, equivalently the same distributional Laplacian

$$-\Delta p = \partial_i \partial_j (u_i u_j),$$

and the de Rham theorem applied to the difference of the two weak formulations gives $\nabla(p - p^L) = 0$ in $\mathcal{D}'(\mathbb{R}^3)$ for a.e. time. Thus $p - p^L$ is spatially constant for a.e. time. Replacing the original pressure by p^L is therefore only a time-dependent gauge change, which does not affect suitability by Theorem 2.5. The rescaled pressure has the same representative with u_k , again modulo a time-dependent function. In the rest of this lemma p_k denotes one such representative; after that representative is fixed, the reference gauge $a_k(t) = (p_k)_{B_1}(t)$ is fixed as well.

Fix a smooth cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on B_{4R} . For $t \in [-S, -\sigma]$, decompose in B_{2R}

$$p_k = q_k + h_k, \quad q_k = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{k,i} u_{k,j}).$$

More precisely, the representative may be chosen, by adding a function of t , so that the difference $h_k = p_k - q_k$ satisfies $-\Delta h_k = 0$ in B_{4R} for a.e. t , because the localized source coincides with $u_{k,i} u_{k,j}$ on B_{4R} . This choice is made before the reference gauge $a_k(t) = (p_k)_{B_1}(t)$ is evaluated, so the subsequent comparison of a_k with $c_{k,R}$ uses a single pressure representative. Calderon–Zygmund boundedness on $L^{3/2}$ gives

$$\|q_k(t)\|_{L^{3/2}(B_{2R})} \leq C \|\chi_R u_k(t) \otimes u_k(t)\|_{L^{3/2}(\mathbb{R}^3)} \leq C \|u_k(t)\|_{L^3(B_{8R})}^2 \leq C(R, \sigma, M).$$

It remains first to control the harmonic far field modulo a constant. Let

$$c_{k,R}(t) = h_k(0, t).$$

The value at the origin is defined by the smooth harmonic representative of $h_k(\cdot, t)$ in B_{4R} ; measurability in t follows by testing the harmonic representative against a fixed mollifier supported in a small ball. Let K_{ij} denote the Calderon–Zygmund kernel of $\mathcal{R}_i \mathcal{R}_j$. For $x \in B_R$, the kernel estimate

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq CR|z|^{-4}, \quad |z| > 4R,$$

implies

$$|h_k(x, t) - c_{k,R}(t)| \leq CR \sum_{m=2}^{\infty} (2^m R)^{-4} \int_{2^m R < |z| < 2^{m+1} R} |1 - \chi_R(z)| |u_k(z, t)|^2 dz.$$

By Theorem 3.1,

$$\int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 dz \leq CM^2 \sigma^{-1} (2^m R)^3.$$

Consequently

$$\|h_k(t) - c_{k,R}(t)\|_{L^\infty(B_R)} \leq CM^2\sigma^{-1}R \sum_{m=2}^{\infty} (2^m R)^{-1} \leq C(M, \sigma).$$

Combining the near-field and far-field estimates and integrating over the finite interval $[-S, -\sigma]$ gives

$$\sup_k \|p_k - c_{k,R}(t)\|_{L^{3/2}(B_R \times [-S, -\sigma])} < \infty.$$

We now pass from the local gauge $c_{k,R}$ to the single reference gauge $a_k(t) = (p_k)_{B_1}(t)$. If $R < 1$, enlarge R to 1 in the preceding estimate; hence assume $R \geq 1$. Since $B_1 \subset B_R$, Jensen's inequality gives, for a.e. $t \in [-S, -\sigma]$,

$$|a_k(t) - c_{k,R}(t)|^{3/2} = |(p_k - c_{k,R}(t))_{B_1}(t)|^{3/2} \leq \frac{1}{|B_1|} \int_{B_1} |p_k - c_{k,R}(t)|^{3/2} dx.$$

Integrating in time shows that $a_k - c_{k,R}$ is uniformly bounded in $L^{3/2}([-S, -\sigma])$. Therefore

$$\|p_k - a_k(t)\|_{L^{3/2}(B_R \times [-S, -\sigma])} \leq \|p_k - c_{k,R}(t)\|_{L^{3/2}} + |B_R|^{2/3} \|c_{k,R} - a_k\|_{L_t^{3/2}},$$

which proves the asserted estimate for the fixed gauge a_k . $\square$

Lemma 3.4 (Time derivative bound). *For every compact cylinder $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$, the sequence $\partial_t u_k$ is bounded in*

$$L_t^{3/2} W_x^{-1,3/2}(Q).$$

Proof. From the equation,

$$\partial_t u_k = \Delta u_k - \operatorname{div}(u_k \otimes u_k) - \nabla(p_k - a_k(t)),$$

because $\nabla a_k(t) = 0$. Let $Q = B_R \times [-S, -\sigma]$. The term Δu_k is bounded in $L_t^2 H_x^{-1}(Q)$ by [Theorem 3.2](#): for test fields $\psi \in L_t^2 H_{0,x}^1(Q)$,

$$\left| \iint \nabla u_k : \nabla \psi \right| \leq \|\nabla u_k\|_{L^2(Q)} \|\nabla \psi\|_{L^2(Q)}.$$

Because $W_0^{1,3}(B_R) \subset H_0^1(B_R)$, an $H^{-1}(B_R)$ -bound implies a $W^{-1,3/2}(B_R)$ -bound. Since the time interval has finite length, the $L_t^2 H_x^{-1}$ -bound is also an $L_t^{3/2} W_x^{-1,3/2}$ -bound. The nonlinear term is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$ because, for every $\zeta \in W_0^{1,3}(B_R)$,

$$\left| \int_{B_R} (u_k \otimes u_k) : \nabla \zeta \, dx \right| \leq \|u_k \otimes u_k\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)},$$

and

$$\|u_k \otimes u_k\|_{L^{3/2}(Q)} \leq \|u_k\|_{L^\infty(Q)} \|u_k\|_{L^{3/2}(Q)} \leq C(Q),$$

using [Theorem 3.1](#), the local L^2 -bound, and the finite measure of Q . Finally, for the pressure term, the dual definition of $W^{-1,3/2}$ gives, for every $\zeta \in W_0^{1,3}(B_R)$,

$$\left| \int_{B_R} (p_k - a_k(t)) \operatorname{div} \zeta \, dx \right| \leq \|p_k - a_k(t)\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)},$$

and hence

$$\|\nabla(p_k - a_k(t))\|_{W^{-1,3/2}(B_R)} \leq \|p_k - a_k(t)\|_{L^{3/2}(B_R)},$$

and [Theorem 3.3](#) gives the required time integrability. $\square$

Proposition 3.5 (Compactness). *After passing to a subsequence,*

$$u_k \rightarrow U$$

strongly in $L^q_{\text{loc}}(\mathbb{R}^3 \times (-\infty, 0))$ for every finite q , and

$$p_k - a_k(t) \rightharpoonup P \quad \text{weakly in } L^{3/2}_{\text{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

Proof. Fix $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$. By [Theorem 3.2](#), the velocities are bounded in $L^2_t H^1_x(Q)$. By [Theorem 3.4](#), the time derivatives are bounded in $L^{3/2}_t W^{-1,3/2}_x(Q)$. On bounded spatial balls,

$$H^1 \Subset L^2 \hookrightarrow W^{-1,3/2}.$$

The Aubin–Lions–Simon theorem [\[2, 7, 11\]](#) therefore gives relative compactness in $L^2(Q)$. After passing to a subsequence, $u_k \rightarrow U$ strongly in $L^2(Q)$ and almost everywhere on Q .

The inherited Type I bound gives a uniform $L^\infty(Q)$ bound. Interpolating between strong $L^2(Q)$ convergence and the common $L^\infty(Q)$ bound gives strong convergence in $L^q(Q)$ for every finite q . The limit belongs to the same $L^\infty(Q)$ ball by weak-star lower semicontinuity in L^∞ , or equivalently by the pointwise a.e. convergence after passing to a subsequence. Thus, for $q \geq 2$,

$$\|u_k - U\|_{L^q(Q)}^q \leq (2 \sup_j \|u_j\|_{L^\infty(Q)} + 2\|U\|_{L^\infty(Q)})^{q-2} \|u_k - U\|_{L^2(Q)}^2,$$

and the case $1 \leq q < 2$ follows from Hölder on the finite-measure set Q . Applying this argument to a countable exhaustion $Q_1 \Subset Q_2 \Subset \dots$ of $\mathbb{R}^3 \times (-\infty, 0)$ by compact cylinders and selecting subsequences successively gives a diagonal subsequence for which the convergence holds on every Q_m , and therefore locally.

Finally, [Theorem 3.3](#) gives uniform local $L^{3/2}$ bounds for the gauged pressures $p_k - a_k(t)$. Since $L^{3/2}$ is reflexive, the Banach–Alaoglu theorem and the same diagonal extraction give a weak $L^{3/2}_{\text{loc}}$ pressure limit. $\square$

4. THE ANCIENT SUITABLE WEAK LIMIT

Lemma 4.1 (Limit equation). *The limit (U, P) obtained in [Theorem 3.5](#) solves*

$$\partial_t U + (U \cdot \nabla)U + \nabla P = \Delta U, \quad \operatorname{div} U = 0$$

in distributions on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. Let $\varphi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0); \mathbb{R}^3)$. The weak formulation for $(u_k, p_k - a_k(t))$ is

$$\iint [-u_k \cdot \partial_t \varphi - (u_k \otimes u_k) : \nabla \varphi + (p_k - a_k(t)) \operatorname{div} \varphi + \nabla u_k : \nabla \varphi] \, dx \, dt = 0.$$

The linear term with $\partial_t \varphi$ passes to the limit by strong L^2_{loc} -convergence of u_k . The pressure term passes to the limit by weak $L^{3/2}_{\text{loc}}$ -convergence of $p_k - a_k(t)$, since $\operatorname{div} \varphi \in L^3$. The viscous term passes to the limit by weak L^2_{loc} -convergence of ∇u_k . The nonlinear term converges because $u_k \rightarrow U$ strongly in L^2_{loc} , hence

$$\|u_k \otimes u_k - U \otimes U\|_{L^1(\operatorname{supp} \varphi)} \leq \|u_k - U\|_{L^2} (\|u_k\|_{L^2} + \|U\|_{L^2}) \rightarrow 0.$$

Testing with gradients of scalar functions gives $\operatorname{div} U = 0$, since $\operatorname{div} u_k = 0$ and $u_k \rightharpoonup U$ in L^2_{loc} . $\square$

Lemma 4.2 (Stability of suitability). *The pair (U, P) satisfies the local energy inequality. Hence it is a suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$.*

Proof. We use the distributional form of the local energy inequality. Namely, for every non-negative $\phi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0))$,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 (\partial_t \phi + \Delta \phi) + \iint (|u_k|^2 + 2P_k) u_k \cdot \nabla \phi,$$

where $P_k = p_k - a_k(t)$. This formulation is obtained from (2.1) by applying the endpoint inequality to time cutoffs and then approximating the temporal cutoff monotonically; conversely it implies the endpoint formulation for the weakly L_{loc}^2 -continuous representative. Thus no additional time-slice convergence is needed in the compactness argument.

The terms involving $|u_k|^2 (\partial_t \phi + \Delta \phi)$ converge strongly in L^1 because $u_k \rightarrow U$ strongly in L_{loc}^2 . The cubic flux term converges strongly in L^1 because $u_k \rightarrow U$ strongly in L_{loc}^3 .

For the pressure flux, since $P_k \rightarrow P$ in $L^{3/2}$ and

$$u_k \cdot \nabla \phi \rightarrow U \cdot \nabla \phi \quad \text{strongly in } L^3,$$

the duality pairing $\iint P_k u_k \cdot \nabla \phi$ converges to $\iint P U \cdot \nabla \phi$. Finally, $\nabla u_k \rightarrow \nabla U$ in L_{loc}^2 , so the convex functional

$$w \mapsto \iint |w|^2 \phi$$

is weakly lower semicontinuous. Taking the lower limit on the left-hand side and the ordinary limits on the right-hand side gives the local energy inequality for U in distributional form. The endpoint form in Theorem 2.2 follows by the same monotone temporal-cutoff approximation and the standard weak L_{loc}^2 -continuity of local energy-class weak solutions. The distributional equation, divergence-free condition, and local integrability requirements were already obtained in Theorems 3.5 and 4.1; hence (U, P) is suitable. $\square$

Lemma 4.3 (Inherited ancient Type I bound). *For every compact interval $I = [-S, -\sigma] \Subset (-\infty, 0)$,*

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M\sigma^{-1/2}.$$

Proof. Let $I = [-S, -\sigma] \Subset (-\infty, 0)$. By Theorem 3.1,

$$\|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M\sigma^{-1/2}.$$

Fix $R < \infty$. The convergence in Theorem 3.5 gives, after passing to a subsequence if necessary, $u_k \rightarrow U$ a.e. on $B_R \times I$. If a measurable set $E \subset B_R \times I$ had positive measure and $|U| > M\sigma^{-1/2} + \delta$ on E , then by a.e. convergence the functions $|u_k|$ would exceed $M\sigma^{-1/2} + \delta/2$ on a subset of E of positive measure for all sufficiently large k , contradicting the uniform bound for u_k . Hence

$$\|U\|_{L^\infty(B_R \times I)} \leq M\sigma^{-1/2}.$$

Letting $R \rightarrow \infty$ through integers gives the asserted bound on $\mathbb{R}^3 \times I$. $\square$

Proposition 4.4 (Ancient suitable Type I limit). *The pair (U, P) is an ancient suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$ satisfying the interval Type I bound in Theorem 4.3.*

Proof. Combine Theorems 4.1 to 4.3. $\square$

5. NON-TRIVIALITY

We now choose the scales so that the ancient limit is not zero. This is the only selection step in the proof, and we separate the local regularity input from the scale selection.

Lemma 5.1 (Pressure-normalized concentration at a singular point). *If $z_* = (x_*, T)$ is singular, then there are $\eta > 0$ and $r_0 > 0$ such that, for every $0 < r < r_0$,*

$$(5.1) \quad C(u; z_*, r) + D(p; z_*, r) \geq \eta.$$

Proof. The existence of a positive CKN concentration sequence at a singular point is proved in the first-node paper [8, Theorem `thm:singular-positive`]. The present stronger all-sufficiently-small-radii form follows directly from the same epsilon-regularity contrapositive, using [8, Theorem `thm:ckn`]. Take $\eta = \varepsilon_*$. If there were arbitrarily small radii r with

$$C(u; z_*, r) + D(p; z_*, r) < \varepsilon_*,$$

then Theorem 2.3 would imply that u is bounded in $Q_{r/2}(z_*)$. This is precisely regularity of z_* in Theorem 2.1, contradicting the hypothesis that z_* is singular. Hence the lower bound holds for all sufficiently small r . $\square$

Lemma 5.2 (Endpoint compactness at a centered Type I point). *Assume $\mathcal{I}_*(u, p; z_*) < \infty$. Let $\lambda_k \downarrow 0$, and let (u_k, p_k) be the centered rescalings (3.2). For every $R < \infty$ there are functions of time $b_{k,R}(t)$ such that, after passing to a subsequence,*

$$u_k \rightarrow U \quad \text{strongly in } L^3(Q_{R'}) \quad \text{for every } 0 < R' < R,$$

and

$$p_k - b_{k,R}(t) \rightharpoonup P \quad \text{weakly in } L^{3/2}(Q_R).$$

The limit is suitable in Q_R . A diagonal subsequence gives this conclusion for every finite R .

Proof. By the scale invariance in Theorem 2.4, the hypothesis $\mathcal{I}_*(u, p; z_*) < \infty$ gives, for each fixed R ,

$$\sup_k [A(u_k; 0, R) + C(u_k; 0, R) + D(p_k; 0, R) + E(u_k; 0, R)] < \infty$$

once k is sufficiently large. The pressure term is understood after subtracting the spatial average on B_R , which is the function $b_{k,R}$. The local energy inequality is invariant under this pressure gauge, since the subtracted term is a function of time and u_k is divergence free.

We spell out the compactness. The A -bound gives

$$\sup_k \|u_k\|_{L^\infty(-R^2, 0; L^2(B_R))} < \infty,$$

and the E -bound gives

$$\sup_k \|\nabla u_k\|_{L^2(Q_R)} < \infty.$$

By the Sobolev embedding $H^1(B_R) \hookrightarrow L^6(B_R)$, the E -bound gives $u_k \in L_t^2 L_x^6(Q_R)$. Interpolating

$$\frac{1}{10/3} = \frac{2}{5} \cdot \frac{1}{2} + \frac{3}{5} \cdot \frac{1}{6}$$

on each time slice gives

$$\|u_k(t)\|_{L^{10/3}(B_R)} \leq \|u_k(t)\|_{L^2(B_R)}^{2/5} \|u_k(t)\|_{L^6(B_R)}^{3/5}.$$

Raising to the power $10/3$ and integrating in time yields

$$\int_{-R^2}^0 \|u_k(t)\|_{L^{10/3}(B_R)}^{10/3} dt \leq \left(\sup_{-R^2 < t < 0} \|u_k(t)\|_{L^2(B_R)}^2 \right)^{2/3} \int_{-R^2}^0 \|u_k(t)\|_{L^6(B_R)}^2 dt.$$

Thus the A - and E -bounds imply the energy-class estimate

$$\sup_k \|u_k\|_{L^{10/3}(Q_R)} < \infty.$$

The equation written with the pressure gauge $p_k - b_{k,R}(t)$ gives

$$\partial_t u_k = \Delta u_k - \operatorname{div}(u_k \otimes u_k) - \nabla(p_k - b_{k,R}(t)).$$

The three terms on the right are bounded in negative Sobolev spaces on every strictly smaller cylinder. The term Δu_k is bounded in $L_t^2 H_x^{-1}$, hence in $L_t^{3/2} W_x^{-1,3/2}$ on finite time intervals, by the E -bound and the embedding $W_0^{1,3}(B_{R'}) \subset H_0^1(B_{R'})$. Since $u_k \in L^{10/3}(Q_R)$,

$$u_k \otimes u_k \in L^{5/3}(Q_R) \subset L^{3/2}(Q_{R'})$$

on the finite-measure cylinder $Q_{R'}$, and therefore $\operatorname{div}(u_k \otimes u_k)$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q_{R'})$. The D -bound gives

$$p_k - b_{k,R}(t) \in L^{3/2}(Q_R),$$

so $\nabla(p_k - b_{k,R}(t))$ is bounded in the same negative Sobolev space. Hence, for each $R' < R$, $\partial_t u_k$ is bounded in

$$L^{3/2}((-R'^2, 0); W^{-1,3/2}(B_{R'})).$$

Since $H^1(B_{R'}) \Subset L^2(B_{R'}) \hookrightarrow W^{-1,3/2}(B_{R'})$, the Aubin–Lions–Simon theorem [2, 7, 11] gives strong convergence in $L^2(Q_{R'})$. The $L^{10/3}$ -bound then upgrades this to strong convergence in $L^3(Q_{R'})$ by interpolation:

$$\|u_k - U\|_{L^3(Q_{R'})} \leq \|u_k - U\|_{L^2(Q_{R'})}^\theta \|u_k - U\|_{L^{10/3}(Q_{R'})}^{1-\theta} \rightarrow 0$$

for the unique $\theta \in (0, 1)$ satisfying $1/3 = \theta/2 + (1 - \theta)/(10/3)$. The D -bound is precisely the $L^{3/2}(Q_R)$ -bound for $p_k - b_{k,R}(t)$, so reflexivity gives a weak $L^{3/2}$ -limit.

It remains to record why the limiting pair is suitable in Q_R . Let $\phi \geq 0$ be compactly supported in Q_R . Apply the local energy inequality to $(u_k, p_k - b_{k,R}(t))$. The terms with $|u_k|^2(\partial_t \phi + \Delta \phi)$ converge because $u_k \rightarrow U$ strongly in L^2 ; the cubic flux converges because $u_k \rightarrow U$ strongly in L^3 ; and the pressure flux converges by the duality between weak $L^{3/2}$ -convergence of $p_k - b_{k,R}(t)$ and strong L^3 -convergence of $u_k \cdot \nabla \phi$. The dissipation term is lower semicontinuous under weak L^2 -convergence of ∇u_k . Thus the local energy inequality passes to the limit. The distributional equation and the divergence-free constraint pass to the limit by the same strong–weak convergences used above. Hence the limit is suitable in Q_R . A diagonal subsequence over $R = 1, 2, \dots$ gives the assertion for all finite R . $\square$

Lemma 5.3 (Scale selection from local concentration). *The scales in Theorem 1.1 may be chosen from a positive local concentration sequence. More precisely, there are $\lambda_k \downarrow 0$, a compact cylinder*

$$K_0 \Subset \mathbb{R}^3 \times (-\infty, 0),$$

the reference gauges $a_k(t) = (p_k)_{B_1}(t)$ from Theorem 3.3, and a number $c_0 > 0$ such that the centered rescalings (3.2) satisfy

$$(5.2) \quad \iint_{K_0} \left(|u_k|^3 + |p_k - a_k(t)|^{3/2} \right) dx dt \geq c_0$$

for all k . In addition, the compactness limit obtained from this selected sequence is nonzero in $L_{\text{loc}}^3(\mathbb{R}^3 \times (-\infty, 0))$.

Proof. By Theorem 5.1, and equivalently by the positive local-concentration theorem in the first-node paper [8, Theorem thm:singular-positive], every sufficiently small radius carries positive pressure-normalized density. Choose any sequence $r_j \downarrow 0$ inside that range, so that

$$C(u; z_*, r_j) + D(p; z_*, r_j) \geq \eta.$$

After rescaling by r_j , this says that the pressure-normalized density of the pair

$$v_j(x, t) = r_j u(x_* + r_j x, T + r_j^2 t), \quad q_j(x, t) = r_j^2 p(x_* + r_j x, T + r_j^2 t)$$

on Q_1 is bounded below by η , with q_j normalized by subtracting $(q_j)_{B_1}(t)$. The subtraction is a legitimate pressure gauge by [Theorem 2.5](#).

We now invoke the Caffarelli–Kohn–Nirenberg–Seregin concentration compactness selection for singular points [3, 5, 9]. In the local Type I setting this is the forward implication in Albritton–Barker [1, Theorem 1.1]: a positive pressure-normalized concentration sequence at a Type I singular terminal point has a subsequence, possibly after replacing the radii by comparable radii $\lambda_j \downarrow 0$, whose rescalings converge to a nonzero bounded ancient limit. The pressures in that compactness statement are normalized by a fixed spatial average, and after the whole-space representative has been chosen as in [Theorem 3.3](#) we take this average to be the reference gauge $a_j(t) = (p_j)_{B_1}(t)$. This is a gauge choice only, by [Theorem 2.5](#), and it is compatible with the local pressure compactness used in [Theorem 3.5](#). In the notation used here, the nonzero conclusion is precisely that a fixed compact subcylinder $K_0 \Subset Q_1 \cap \{t < 0\}$ carries a uniform positive amount of pressure-normalized density. Thus there are K_0 and $c_0 > 0$ such that

$$\iint_{K_0} \left(|u_j|^3 + |p_j - a_j(t)|^{3/2} \right) dx dt \geq c_0.$$

Here u_j, p_j denote the centered rescalings at the selected scales λ_j , and $a_j(t) = (p_j)_{B_1}(t)$.

No additional hypothesis is being imposed in this selection step. The input is only the positive CKN concentration supplied by [Theorem 5.1](#), the local Type I alternative from [8, Theorem [thm:blowup-rate-dichotomy](#)], and the endpoint compactness supplied by $\mathcal{I}_*(u, p; z_*) < \infty$. The possible loss of density into the terminal slice is the defect excluded by the cited Seregin selection theorem.

The global Type I bound (1.2) rules out the Type II branch of the local blow-up-rate dichotomy proved in the first-node paper [8, Theorem [thm:blowup-rate-dichotomy](#)] along the selected sequence, while $\mathcal{I}_*(u, p; z_*) < \infty$ gives the endpoint compactness needed above. The selected sequence is therefore a sequence for which the compactness limit is nonzero. Renaming that selected sequence as k gives (5.2) and the final assertion of the lemma. $\square$

Remark 5.4. The scale-selection lemma is the only point at which singularity, rather than mere bounded Type I behavior, is used. It is also the point at which the terminal-slice concentration defect is excluded. For this reason the lower bound (5.2) is placed on a cylinder compactly contained in $\{t < 0\}$.

Proposition 5.5 (Nonzero ancient limit). *For the scales chosen in [Theorem 5.3](#), the limit U in [Theorem 4.4](#) is nonzero. More precisely, there are a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, a time-dependent pressure gauge $a(t)$, and a number $\eta_1 > 0$ such that*

$$\iint_{K_0} \left(|U|^3 + |P - a(t)|^{3/2} \right) dx dt \geq \eta_1.$$

Proof. The scale-selection theorem invoked in [Theorem 5.3](#) is used here with its nonzero-limit conclusion. We apply the compactness construction of [Theorem 3.5](#) to that already selected sequence. If a further subsequence is needed for local compactness, it is taken from a sequence which already converges, in the cited selection theorem, to a bounded ancient suitable weak solution which is not identically zero. The further subsequence therefore has the same limit. Thus the limit denoted by U in [Theorem 4.4](#) satisfies

$$U \not\equiv 0 \quad \text{in } L^3_{\text{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

Since $U \in L^3_{\text{loc}}$ and is not zero, there is a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$ such that

$$\iint_{K_0} |U|^3 dx dt > 0.$$

Let $a(t)$ be any time-dependent pressure gauge for P on K_0 , for instance the spatial average of P over a ball containing the spatial projection of K_0 . Then

$$\iint_{K_0} (|U|^3 + |P - a(t)|^{3/2}) dx dt \geq \iint_{K_0} |U|^3 dx dt.$$

Taking η_1 to be the positive right-hand side gives the asserted pressure-normalized lower bound. No lower semicontinuity of the pressure norm is used in this step. $\square$

6. CENTERED SELF-SIMILAR VARIABLES

The Type I bound makes the following change of variables natural:

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t).$$

Set also

$$\Pi(y, \tau) = (-t)P(y\sqrt{-t}, t),$$

up to addition of a function of τ .

Lemma 6.1 (Renormalized equation). *The pair (V, Π) solves*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0.$$

Proof. Write $s = -t = e^{-\tau}$, $x = s^{1/2}y$, and

$$U(x, t) = s^{-1/2}V(y, \tau).$$

The identities below are identities in distributions obtained by testing against the inverse change of variables. Since [Theorem 6.3](#) gives smooth representatives on compact subsets of $t < 0$, the same computation is classical for the representative appearing in the theorem. Since $s_t = -1$, $\tau_t = s^{-1}$, and $y_t = (2s)^{-1}y$, one computes

$$\begin{aligned} \partial_t U &= s^{-3/2} \left(\partial_\tau V + \frac{1}{2}V + \frac{1}{2}y \cdot \nabla V \right), \\ (U \cdot \nabla_x)U &= s^{-3/2}(V \cdot \nabla)V, \quad \Delta_x U = s^{-3/2}\Delta V, \quad \nabla_x P = s^{-3/2}\nabla \Pi. \end{aligned}$$

Substitution in the Navier–Stokes equation for U gives the displayed equation. The divergence condition is preserved by the spatial scaling. $\square$

Lemma 6.2 (Renormalized boundedness and non-triviality). *The field V satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M.$$

Moreover [\(1.8\)](#) *holds for some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, function $b(\tau)$, and $\eta_0 > 0$.*

Proof. The L^∞ bound follows from the classical pointwise Type I bound in [Theorem 6.4](#):

$$|V(y, \tau)| = \sqrt{-t} |U(y\sqrt{-t}, t)| \leq M.$$

For non-triviality, [Theorem 5.5](#) gives positive pressure-normalized CKN density on a compact spacetime cylinder in (x, t) . Under the change of variables $(x, t) \mapsto (y, \tau)$, this compact cylinder maps into a compact set contained in some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$. If $s = -t = e^{-\tau}$, then

$$dx dt = s^{5/2} dy d\tau, \quad U = s^{-1/2}V, \quad P - a(t) = s^{-1}(\Pi - b(\tau)),$$

where $b(\tau) = s a(-s)$. Therefore

$$|U|^3 dx dt = s |V|^3 dy d\tau, \quad |P - a(t)|^{3/2} dx dt = s |\Pi - b(\tau)|^{3/2} dy d\tau.$$

On the image of the compact cylinder, s is bounded above and below by positive constants. Thus a positive lower bound in (x, t) -variables is equivalent, up to a fixed multiplicative constant, to a positive lower bound in (y, τ) -variables. This gives (1.8) with a possibly different positive constant η_0 . $\square$

Lemma 6.3 (Smoothness). *The ancient solution U is smooth on $\mathbb{R}^3 \times (-\infty, 0)$, and V is smooth on $\mathbb{R}^3 \times \mathbb{R}$.*

Proof. On each compact cylinder $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$, the bound Theorem 4.3 gives $U \in L^\infty(Q)$. In particular $U \in L_t^s L_x^r(Q)$ for every finite r, s . Choose, for example, $r = s = 10$; then $2/s + 3/r = 1/2 < 1$. Serrin's interior regularity criterion [10, 12] for suitable weak solutions applies on Q . In the form used here, the criterion asserts that every point of Q is regular: for each $Q' \Subset Q$, U is bounded and Hölder continuous on Q' , and the interior bootstrap for the Stokes system gives smoothness on still smaller cylinders. The bootstrap starts from the equation

$$\partial_t U - \Delta U + \nabla P = -\operatorname{div}(U \otimes U), \quad \operatorname{div} U = 0,$$

with $U \otimes U$ locally bounded after Serrin regularity. Interior Stokes estimates on nested cylinders give higher spatial derivatives and one time derivative; differentiating the equation and iterating the same estimates gives

$$U \in C^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

The pressure is recovered locally from $-\Delta P = \partial_i \partial_j (U_i U_j)$ plus a harmonic function and is smooth after fixing a time-dependent gauge. The self-similar change of variables is a smooth diffeomorphism for $t < 0$, so V and Π are smooth on $\mathbb{R}^3 \times \mathbb{R}$. $\square$

Corollary 6.4 (Classical Type I bound). *For the smooth representative from Theorem 6.3,*

$$\|U(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}, \quad t < 0.$$

Proof. Fix $t_0 < 0$. Choose $\delta_n \downarrow 0$ such that

$$I_n = [t_0 - \delta_n, t_0 + \delta_n] \Subset (-\infty, 0).$$

By Theorem 4.3,

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I_n)} \leq M(-t_0 - \delta_n)^{-1/2}.$$

Since the representative is continuous in time and space, this essential spacetime bound holds pointwise on $\mathbb{R}^3 \times I_n$: otherwise a point where the bound is exceeded would have a small spacetime neighborhood of positive measure where it is still exceeded. Hence, for every $x \in \mathbb{R}^3$,

$$|U(x, t_0)| \leq M(-t_0 - \delta_n)^{-1/2} \quad \text{for every } n.$$

Letting $n \rightarrow \infty$ and then taking the supremum over $x \in \mathbb{R}^3$ gives the asserted estimate. $\square$

Proof of Theorem 1.1. The compact ancient limit and the interval Type I bound are Theorem 4.4; the pointwise bound (1.7) is Theorem 6.4. Non-triviality is Theorem 5.5. The renormalized equation is Theorem 6.1, and the boundedness and renormalized non-triviality are Theorem 6.2. Smoothness is Theorem 6.3. $\square$

7. ABSENCE OF ADDITIONAL CENTERED SYMMETRIES

Lemma 7.1 (Residual logarithmic-time translation). *If V solves (1.5), then for every $\tau_0 \in \mathbb{R}$ the function*

$$V_{\tau_0}(y, \tau) = V(y, \tau + \tau_0)$$

also solves (1.5). Moreover, if

$$U^\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t)$$

is a parabolic rescaling of the unrenormalized ancient solution, then the renormalized field associated with U^λ is

$$V^\lambda(y, \tau) = V(y, \tau - 2 \log \lambda).$$

Proof. The first statement follows because (1.5) is autonomous in τ . For the second, write $s = -t = e^{-\tau}$. Then

$$\sqrt{s} U^\lambda(y\sqrt{s}, -s) = \lambda\sqrt{s} U(\lambda y\sqrt{s}, -\lambda^2 s) = V(y, -\log(\lambda^2 s)) = V(y, \tau - 2 \log \lambda).$$

□

Lemma 7.2 (Failure of scaling invariance). *Let V be a bounded smooth solution of (1.5). For $\alpha > 0$, define*

$$\tilde{V}(y, \tau) = \alpha V(\alpha y, \alpha^2 \tau).$$

If $V \not\equiv 0$ and $\tilde{V}$ solves the same centered equation with the pressure $\tilde{\Pi}(y, \tau) = \alpha^2 \Pi(\alpha y, \alpha^2 \tau)$, then $\alpha = 1$.

Proof. Write $Y = \alpha y$ and $T = \alpha^2 \tau$. The terms

$$\partial_\tau V, \quad (V \cdot \nabla)V, \quad \nabla \Pi, \quad \Delta V$$

in the non-drift part scale with the common factor α^3 . More precisely,

$$\partial_\tau \tilde{V} + (\tilde{V} \cdot \nabla) \tilde{V} + \nabla \tilde{\Pi} - \Delta \tilde{V} = \alpha^3 (\partial_T V + (V \cdot \nabla_Y)V + \nabla_Y \Pi - \Delta_Y V)(Y, T).$$

The drift term satisfies

$$-\frac{1}{2}(\tilde{V} + y \cdot \nabla \tilde{V}) = -\frac{\alpha}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

Using the equation for V , the residual of the centered equation for $\tilde{V}$ is therefore

$$\frac{\alpha - \alpha^3}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

If $\tilde{V}$ solves the equation and $\alpha \neq 1$, then

$$V(Y, T) + Y \cdot \nabla_Y V(Y, T) = 0 \quad \text{for all } (Y, T).$$

Fix T and a direction $\omega \in S^2$. For $r > 0$,

$$\frac{d}{dr}(rV(r\omega, T)) = V(r\omega, T) + r\omega \cdot \nabla V(r\omega, T) = 0.$$

Hence $rV(r\omega, T)$ is constant in r . Since V is bounded and smooth at $Y = 0$, this constant must be zero, and so $V(r\omega, T) = 0$ for all $r > 0$. By continuity, $V(\cdot, T) \equiv 0$, and since T was arbitrary, $V \equiv 0$. This contradicts $V \not\equiv 0$; hence $\alpha = 1$. □

Lemma 7.3 (Failure of translation invariance). *Let $a \in \mathbb{R}^3$, and define $\tilde{V}(y, \tau) = V(y - a, \tau)$. The map $V \mapsto \tilde{V}$ is a symmetry of the centered equation only when $a = 0$. For a fixed solution V , the translated field can solve the same equation only if $a \cdot \nabla V \equiv 0$.*

Proof. All translation-invariant terms in the equation transform by composition with the shifted variable. The drift does not:

$$y \cdot \nabla \tilde{V}(y, \tau) = y \cdot \nabla V(y - a, \tau) = (y - a) \cdot \nabla V(y - a, \tau) + a \cdot \nabla V(y - a, \tau).$$

The extra term $a \cdot \nabla V(y - a, \tau)$ is absent from the original equation. Hence translation invariance fails unless $a = 0$. For a fixed solution, the translated field can still solve the same equation only in the exceptional case $a \cdot \nabla V \equiv 0$. $\square$

Proof of Theorem 1.2. Theorem 7.1 identifies the residual action of the original parabolic scaling: it is translation in logarithmic time. Theorem 7.2 excludes any additional non-trivial amplitude-space scaling symmetry, and Theorem 7.3 shows that constant spatial recentering is not a symmetry of the fixed centered equation. Therefore any multi-scale behavior of a bounded ancient profile must be encoded in the evolution of that profile in τ , not in an orbit generated by further centered rescalings. $\square$

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